

To prove (5.1), we make an elementary algebraic re-ordering of the left side of (5.1) into five summands and invoke the triangle inequality successively:

$$\begin{aligned}
(5.1), \text{ left side} &= \|(A - \lambda)^{-1}(-\Delta A)(A + \Delta A - \lambda)^{-1} \\
&\quad - \text{Op}\left(\psi(a_m - \lambda)^{-1}(-\Delta a_m)(a_m + \Delta a_m - \lambda)^{-1}\right)\|_{s,s+m} \\
&\leq \|(A - \lambda)^{-1} - \text{Op}(\chi(a_m - \lambda)^{-1})\|_{s,s+m} \\
&\quad \cdot \|\Delta A(A + \Delta A - \lambda)^{-1}\|_{s,s} \\
&\quad + \|\text{Op}(\chi(a_m - \lambda)^{-1})\|_{s,s+m} \cdot \\
&\quad \|\Delta A - \text{Op}(\chi_1 \Delta a_m)\|_{s+m-1,s} \|(A + \Delta A - \lambda)^{-1}\|_{s,s+m-1} \\
&\quad + \|\text{Op}(\chi(a_m - \lambda)^{-1}) \text{Op}(\chi_1 \Delta a_m)\|_{s+m,s+m} \cdot \\
&\quad \|A + \Delta A - \lambda)^{-1} - \text{Op}(\chi_2(a_m + \Delta a_m - \lambda)^{-1})\|_{s,s+m} \\
&\quad + \|\left\{ \text{Op}(\chi(a_m - \lambda)^{-1}) \text{Op}(\chi_1 \Delta a_m) - \text{Op}(\chi(a_m - \lambda)^{-1} \Delta a_m) \right\} \\
&\quad \cdot \text{Op}(\chi_2(a_m + \Delta a_m - \lambda)^{-1})\|_{s,s+m} \\
&\quad + \|\text{Op}(\chi(a_m - \lambda)^{-1} \Delta a_m) \text{Op}(\chi_2(a_m + \Delta a_m - \lambda)^{-1}) \\
&\quad - \text{Op}(\psi(a_m - \lambda)^{-1} \Delta a_m (a_m + \Delta a_m - \lambda)^{-1})\|_{s,s+m}.
\end{aligned}$$

Here we choose χ, χ_1, χ_2 with the same properties like ψ such that $\chi = \chi\chi_1$ and $\chi\chi_2 = \psi$.

Now apply the Technical Lemma 4.2 to the last two summands and the Lemmata 4.7 and 4.8 to the first three summands, and we are done.

Remark 5.1. We note in passing that in the first three lines of the proof of Proposition 3.1 it is decisively used that the integrand is *homogeneous* in λ . This is an obstruction against an immediate generalization of our perturbation result to general holomorphic functions of A in an H_∞ -calculus style (cf., e.g., [7]). To explain this a bit more, let f be a function, which is holomorphic and bounded in a neighborhood of the closure of the sector encircled by Γ_+ . This is not quite the situation of the H_∞ calculus, since there are also eigenvalues on the left of the contour. Nevertheless, one might hope that similarly to loc. cit. one can prove that $f(A)$ is a bounded operator. Still there is no immediate analogue of Proposition 3.1 in this case and we leave it as an intriguing open problem whether the Perturbation Theorem 1.2 carries over to $f(A)$ instead of $P_{\Gamma_+}(A)$.

5.3. Index correction formulas. The sectorial projections are significant in the celebrated Atiyah-Patodi-Singer Index Theorem. A common set-up is the following: Let X be a compact smooth Riemannian manifold with boundary M , and E and F be two Hermitian vector bundles over X . Let $D : H^1(X; E) \rightarrow L^2(X; F)$ be a first order elliptic differential operator and let $A : H^1(M; E|_M) \rightarrow L^2(M; E|_M)$ denote the tangential operator of D on M relative to the fixed metric structures. In the classical works [2, 3, 4], M. ATIYAH, V. PATODI AND I. SINGER assumed that D is of Dirac type;